

Academic CV

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Associate Professor, Research Center for Eco-Environmental Sciences, Chinese Academy of Sciences

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Education and appointments

- Associate Professor, Research Center for Eco-Environmental Sciences, Chinese Academy of Sciences; State Key Laboratory of Regional and Urban Ecology (SKLRUE), formerly State Key Laboratory of Urban and Regional Ecology — April 2017 – present
- Postdoctoral Fellow (Ecological Remote Sensing), Aerospace Information Research Institute, Chinese Academy of Sciences (AIRCAS), formerly RADIS — August 2014 – April 2017
- Ph.D., Remote Sensing, Hokkaido University — October 2010 – June 2014. Thesis: 遥感卫星观测及模型在生物质火烧中的应用研究
- M.Sc., Physical Geography, Northeast Normal University — September 2008 – July 2010
- B.Sc., Geography, Harbin College — September 2003 – July 2007

Research

Research direction: 生态环境遥感研究.

Research addresses global climate change and eco-environmental remote sensing. Typical ecosystems—wetlands, forests and grasslands—in cities, nature reserves and national parks are examined in relation to drought, flood, wildfire and other extreme meteorological hazards. Threatened migratory birds are treated as indicator species for migration mechanisms, flyway-corridor conservation, and responses to human activity and extreme climate events. Applications include Sino BON, restoration of typical degraded wetlands in Northeast China, bird-habitat planning for the Xiong'an New Area, Nanchang territorial spatial planning (2019–2035), and a bird-strike risk early-warning system. Public academic identity: Movement Ecology × Earth Observation × Biodiversity Conservation.

Hosted projects

- 鸟类多尺度运动行为模式及其迁徙路线核心栖息地监测与保护研究. NSFC General Programme, 2023.01–2026.12, Principal investigator
- 基于多源遥感和卫星追踪大数据的鸟类运动行为信息反演研究. NSFC General Programme, 2019.01–2022.12, Principal investigator

Also hosted a National Key R&D; Programme sub-project and locally commissioned studies. Output overview: More than 40 academic papers; more than ten hosted projects (National Natural Science Foundation of China, a National Key R&D; Programme sub-project, and local commissions) with total hosted funding exceeding RMB 8 million; three invention patents; two co-authored or edited books.

Selected publications

- Zhang, X., Wan, W., Yang, H. & Yi, K. (2025). Rapid development of wind energy infrastructure threatens bird migration. *The Innovation Geoscience* 3(3): 100142. <https://doi.org/10.59717/j.xinn-geo.2025.100142>
- Tian, Z., Huo, D., Yi, K., Que, J., Lu, Z. & Hou, J. (2024). Evaluation of Suitable Habitats for Birds Based on MaxEnt and Google Earth Engine—A Case Study of Baer’ s Pochard (*Aythya baeri*) in Baiyangdian, China. *Remote Sensing* 16(1): 64. <https://doi.org/10.3390/rs16010064>
- Yi, K., Meng, F., Gu, D. & Miao, Q. (2023). Optimizing Water Level Management Strategies to Strengthen Reservoir Support for Bird’ s Migration Network. *Remote Sensing* 15(23): 5508. <https://doi.org/10.3390/rs15235508>
- Yi, K., Zhang, J., Batbayar, N., Higuchi, H., Natsagdorj, T. & Bysykatova, I. P. (2022). Using Tracking Data to Identify Gaps in Knowledge and Conservation of the Critically Endangered Siberian Crane (*Leucogeranus leucogeranus*). *Remote Sensing* 14(20): 5101. <https://doi.org/10.3390/rs14205101>
- Batbayar, N., Yi, K., Zhang, J., Natsagdorj, T., Damba, I., Cao, L. & Fox, A. D. (2021). Combining Tracking and Remote Sensing to Identify Critical Year-Round Site, Habitat Use and Migratory Connectivity of a Threatened Waterbird Species. *Remote Sensing* 13(20): 4049. <https://doi.org/10.3390/rs13204049>

Publications

- Yu, M., Yin, L. & Yi, K. (2026). Bird diversity responses to mega-events influence: A case study of the Beijing 2022 Winter Olympics. *Ecological Informatics* 93: 103594. <https://doi.org/10.1016/j.ecoinf.2025.103594>
- Zhang, X., Wan, W., Yang, H. & Yi, K. (2025). Rapid development of wind energy infrastructure threatens bird migration. *The Innovation Geoscience* 3(3): 100142. <https://doi.org/10.59717/j.xinn-geo.2025.100142>
- Yin, S., Yi, K., Zhang, X., Nie, T., Meng, L., Sun, Z., Chu, Q., Ai, Z., Zhao, X., Wu, L., Guo, M. & Liu, X. (2024). Temporal and Spatial Dynamics of Summer Crop Residue Burning Practices in North China. *Remote Sensing* 16(24): 4763. <https://doi.org/10.3390/rs16244763>
- Tian, Z., Huo, D., Yi, K., Que, J., Lu, Z. & Hou, J. (2024). Evaluation of Suitable Habitats for Birds Based on MaxEnt and Google Earth Engine—A Case Study of Baer’ s Pochard (*Aythya baeri*) in Baiyangdian, China. *Remote Sensing* 16(1): 64. <https://doi.org/10.3390/rs16010064>
- Yi, K., Meng, F., Gu, D. & Miao, Q. (2023). Optimizing Water Level Management Strategies to Strengthen Reservoir Support for Bird’ s Migration Network. *Remote Sensing* 15(23): 5508. <https://doi.org/10.3390/rs15235508>
- Yi, K., Zhao, X., Zheng, Z., Zhao, D. & Zeng, Y. (2023). Trends of greening and browning in terrestrial vegetation in China from 2000 to 2020. *Ecological Indicators* 154: 110587. <https://doi.org/10.1016/j.ecolind.2023.110587>
- Yi, K., Zhang, J., Batbayar, N., Higuchi, H., Natsagdorj, T. & Bysykatova, I. P. (2022). Using Tracking Data to Identify Gaps in Knowledge and Conservation of the Critically Endangered Siberian Crane (*Leucogeranus leucogeranus*). *Remote Sensing* 14(20): 5101. <https://doi.org/10.3390/rs14205101>
- Bao, Y., Shinoda, M., Yi, K., et al. (2022). Satellite-Based Analysis of Spatiotemporal Wildfire Pattern in the Mongolian Plateau. *Remote Sensing* 15(1): 190. <https://doi.org/10.3390/rs15010190>
- Ren, X., Yi, K. & Cao, L. (2022). 优化风电场和电网布局以避免鸟撞. *环境保护科学* 48(3): 13–19.
- Batbayar, N., Yi, K., Zhang, J., Natsagdorj, T., Damba, I., Cao, L. & Fox, A. D. (2021). Combining Tracking and Remote Sensing to Identify Critical Year-Round Site, Habitat Use and Migratory Connectivity of a Threatened Waterbird Species. *Remote Sensing* 13(20): 4049. <https://doi.org/10.3390/rs13204049>

- Damba, I., Zhang, J., Yi, K., et al. (2021). Seasonal and regional differences in migration patterns and conservation status of Swan Geese (*Anser cygnoides*) in the East Asian Flyway. *Avian Research* 12(1): 73. <https://doi.org/10.1186/s40657-021-00308-y>
- Li, H., Fang, L., Wang, X., Yi, K., Cao, L. & Fox, A. D. (2020). Does snowmelt constrain spring migration progression in sympatric wintering Arctic-nesting geese? *Ibis* 162(2): 548–555. <https://doi.org/10.1111/ibi.12767>
- Zhu, Q., Damba, I., Zhao, Q., Yi, K., et al. (2020). Lack of conspicuous sex-biased dispersal patterns at different spatial scales in an Asian endemic goose species breeding in unpredictable steppe wetlands. *Ecology and Evolution* 10: 7006–7020. <https://doi.org/10.1002/ece3.6382>
- Damba, I., Fang, L., Yi, K., et al. (2020). Flyway structure, breeding, migration and wintering distributions of the globally threatened Swan Goose *Anser cygnoides* in East Asia. *Wildfowl*: 97–123.
- Yan, M., Yi, K., Zhang, J., et al. (2020). Flyway connectivity and population status of the Greylag Goose *Anser anser* in East Asia. *Wildfowl* 6: 157–180.
- Mao, D., Wang, Z., Wu, J., Wu, B., Zeng, Y., Song, K., Yi, K. & Luo, L. (2018). China's wetlands loss to urban expansion. *Land Degradation & Development* 29(8): 2644–2657. <https://doi.org/10.1002/ldr.2939>
- Wang, X., Cao, L., Bysykatova, I., et al. (2018). The Far East taiga forest: unrecognized inhospitable terrain for migrating Arctic-nesting waterbirds? *PeerJ* 6: e4353. <https://doi.org/10.7717/peerj.4353>
- Yi, K., Bao, Y. & Zhang, J. (2017). Spatial distribution and temporal variability of open fire in China. *International Journal of Wildland Fire* 26(2): 122–135. <https://doi.org/10.1071/WF15213>
- Yi, K., Zeng, Y. & Wu, B. (2016). Mapping and evaluation the process, pattern and potential of urban growth in China. *Applied Geography* 71: 44–55. <https://doi.org/10.1016/j.apgeog.2016.04.011>
- Yi, K. & Bao, Y. (2016). Estimates of Wildfire Emissions in Boreal Forests of China. *Forests* 7(8): 158. <https://doi.org/10.3390/f7080158>
- Yi, K., Tani, H., Li, Q., Zhang, J., Guo, M., Bao, Y., Wang, X. & Li, J. (2014). Mapping and Evaluating the Urbanization Process in Northeast China Using DMSP/OLS Nighttime Light Data. *Sensors* 14(2): 3207–3226. <https://doi.org/10.3390/s140203207>
- Yi, K., Tani, H., Zhang, J., Guo, M., Wang, X. & Zhong, G. (2013). Long-Term Satellite Detection of Post-Fire Vegetation Trends in Boreal Forests of China. *Remote Sensing* 5(12): 6938–6957. <https://doi.org/10.3390/rs5126938>
- Guo, M., Wang, X., Li, J., Yi, K., Zhong, G. & Tani, H. (2012). Assessment of Global Carbon Dioxide Concentration Using MODIS and GOSAT Data. *Sensors* 12(12): 16368–16389. <https://doi.org/10.3390/s121216368>

Patents and books

Three invention patents. Two co-authored or edited books.

Professional service

Council member, China Grassland Society. Member of the Geographical Society of China, the Ecological Society of China, the Chinese Society of Remote Sensing, and the China Ornithological Society. Guest Editor of special issues for *Remote Sensing*, *Land and Fire*.

Students

- Master' s supervisor (导师).
- Directions: 运动生态学, 迁徙动物监测与保护; 湿地监测与生态修复; 生态系统遥感监测与评估.

- Majors: 071300 Ecology; 0705Z1 Natural Resources Science; 070503 Cartography and Geographic Information Systems.
- Prospective students: write to kpyi@rcees.ac.cn.

Contact

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Laboratory English name: State Key Laboratory of Regional and Urban Ecology (SKLRUE), formerly State Key Laboratory of Urban and Regional Ecology. Laboratory website: <https://dse.rcees.cas.cn/>